Understanding Linear Independence in Vector Spaces

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Read Section 2.5: “Linear Independence” from Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this lesson, you will examine the concept of linear independence, which is fundamental to understanding the structure and behavior of vector spaces. You will learn how to determine whether a set of vectors is linearly independent or dependent and explore the significance of this property in defining the dimension and basis of a vector space.

As you read, consider the following questions:

- What criteria must be met for a set of vectors to be considered linearly independent?
- How does the concept of linear independence relate to the dimension of a vector space?
- Why is linear independence important when constructing a basis for a vector space?
- How can identifying linearly independent vectors help in simplifying and solving complex problems in machine learning?

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